

Write Inequalities

Lesson 7-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x > 5$ means x can be 6, 7, 8, ... — any number greater than 5

Inequality

Write the definition:

$8 > 3$ — eight is larger than three

Greater than

Write the definition:

$2 < 7$ — two is smaller than seven

Less than

Write the definition:

'At least 10 players' $\rightarrow p \geq 10$ (could be 10, 11, 12, ...); 'At most 5 tries' $\rightarrow t \leq 5$ (could be 5, 4, 3, ...)

At least / At most

Write the definition:

x

stands for a number

In $x > 5$, the letter x stands for any number greater than 5.

Variable

Write the definition:

#

number

"No more than 8 people" means $\text{people} \leq 8$.

No more than

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write inequalities to represent real-world situations.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Inequality

Greater than

Less than

At least / At most

Variable

No more than

1 A math sentence comparing values with $<$, $>$, $\leq$, or $\geq$ is an _____.

2 The symbol $>$ that means one value is larger than another means _____.

3 The symbol $<$ that means one value is smaller than another means _____.

4 'At least' means greater than or equal to, and 'at most' means _____ than or equal to.

5 A letter that stands for an unknown number is a _____.

6 The phrase that means less than or equal to is "no more than," written with the _____ symbol.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which inequality represents: 'The temperature is less than 40 degrees'?

- A. $t < 40$
- B. $t > 40$
- C. $t \leq 40$
- D. $t = 40$

Step 1 'Less than' means $<$, so the inequality is $t < 40$.

 **Answer:** A. $t < 40$

 We do — together

Solve this one with your class using the same steps.

Problem: Which inequality represents: 'You must be at least 48 inches tall to ride'?

- A. $h \geq 48$
- B. $h > 48$
- C. $h < 48$
- D. $h \leq 48$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which inequality represents: 'A store can hold no more than 75 people'?

- A. $p \leq 75$
- B. $p < 75$
- C. $p \geq 75$
- D. $p > 75$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A clue says the suspect carried at least 12 stolen gems. Which inequality shows the number g of gems?

A. $g \geq 12$

B. $g > 12$

C. $g \leq 12$

D. $g < 12$

Show your work:

2. A witness reports that fewer than 5 people entered the building. Which inequality shows the number n of people?

A. $n < 5$

B. $n \leq 5$

C. $n > 5$

D. $n \geq 5$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write two different real-world situations: one that uses $>$ and one that uses $\geq$.

Explain why the situations need different symbols even though both mean 'bigger.'

Sentence starter: Situation 1 ($>$): _____. This uses $>$ because _____. Situation 2 ($\geq$): _____. This uses $\geq$ because _____. The difference is that $\geq$ includes _____ while $>$ does not.

Show your work:

Reflect — Exit Ticket

Which inequality represents: 'A number n is at least 14'?

- A. $n \geq 14$
- B. $n > 14$
- C. $n \leq 14$
- D. $n < 14$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Inequality
2. **Notes 2:** Greater than
3. **Notes 3:** Less than
4. **Notes 4:** At least / At most
5. **Notes 5:** Variable
6. **Notes 6:** No more than
7. **We Try (notes):** A. $h \geq 48$ — *'At least' means greater than or equal to, so $h \geq 48$.*
8. **You Try (notes):** A. $p \leq 75$ — *'No more than' means less than or equal to, so $p \leq 75$.*
9. **Try It 1:** A. $g \geq 12$ — *'At least 12' includes 12 and any number above it, so $g \geq 12$.*
10. **Try It 2:** A. $n < 5$ — *'Fewer than 5' does not include 5, so $n < 5$.*
11. **Exit Ticket:** A. $n \geq 14$ — *'At least 14' means 14 or greater, so $n \geq 14$.*